

Reinforcement Learning: Policy Gradients: Bias and Variance (ESE 3990)

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Categorizing RL agents

Axis 1: How to make decisions

- Value Based
 - No Policy (Implicit)
 - Value/Q Function
- Policy Based
 - Policy
 - No Value/Q Function
- Actor Critic
 - Policy
 - Value Function

Today

Axis 2: Environment Representation

- Model-based
 - Policy and/or Value/Q Function
 - Model
- Model-free
 - Policy and/or Value/Q Function
 - No Model

Reinforcement Learning by Policy Gradients

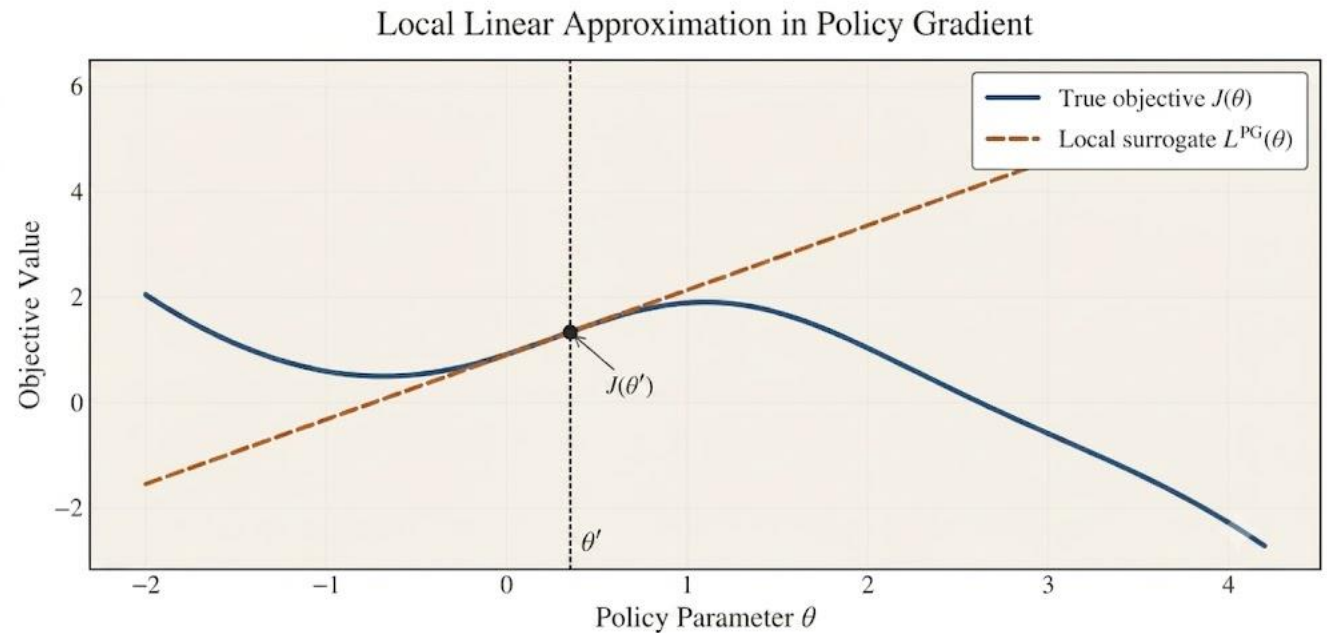
- True Objective

$$J(\theta) = \mathbb{E}_{\tau \sim p(\tau; \theta)} [R(\tau)]$$

- Local Surrogate Objective

- $L(\theta) = \sum_{t=0}^{T-1} (\log \pi(a_t | s_t, \theta) G_t)$

- The **local surrogate** and **true objective** have the same gradient at the parameters θ' of the policy that collected the trajectories.



Reinforce ('92)

- Initialize θ arbitrarily
- Until tired:
 - Run the policy $\pi(a|s, \theta)$ for K episodes of length T
 - Compute policy gradient


$$\hat{\nabla}_{\theta} J(\theta) = \frac{1}{K} \sum_{k=0}^{K-1} \left[\sum_{t=0}^{T-1} \nabla_{\theta} \log \pi(a_t^k | s_t^k, \theta) \textcolor{red}{G}_t^k \right]$$

- $\theta \leftarrow \theta + \alpha \hat{\nabla}_{\theta} J(\theta)$
- Return θ

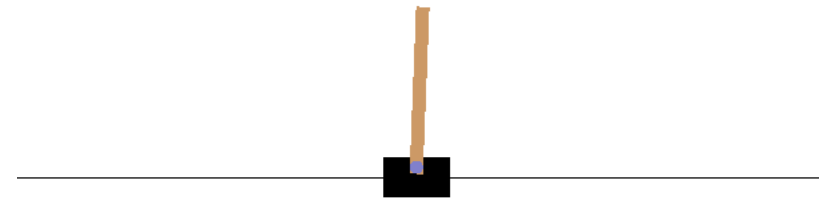
Reinforce: Advantages and Limitations

Advantages

- Extremely easy to use
- Good performance in complex problems (after enough tuning)

Limitations

- High-variance
- Very hard to tune



Example: Bandits

Choose

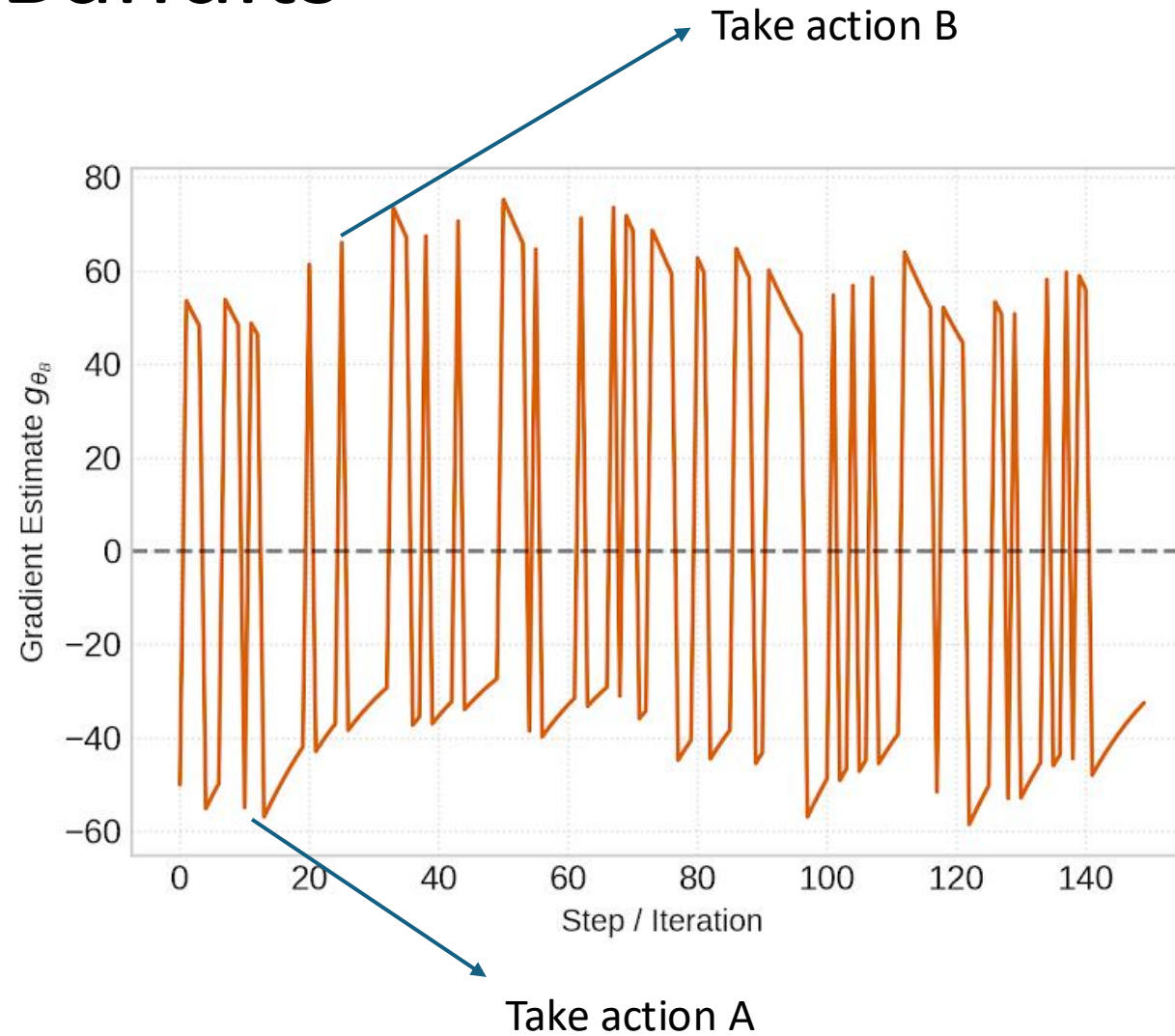
A

B

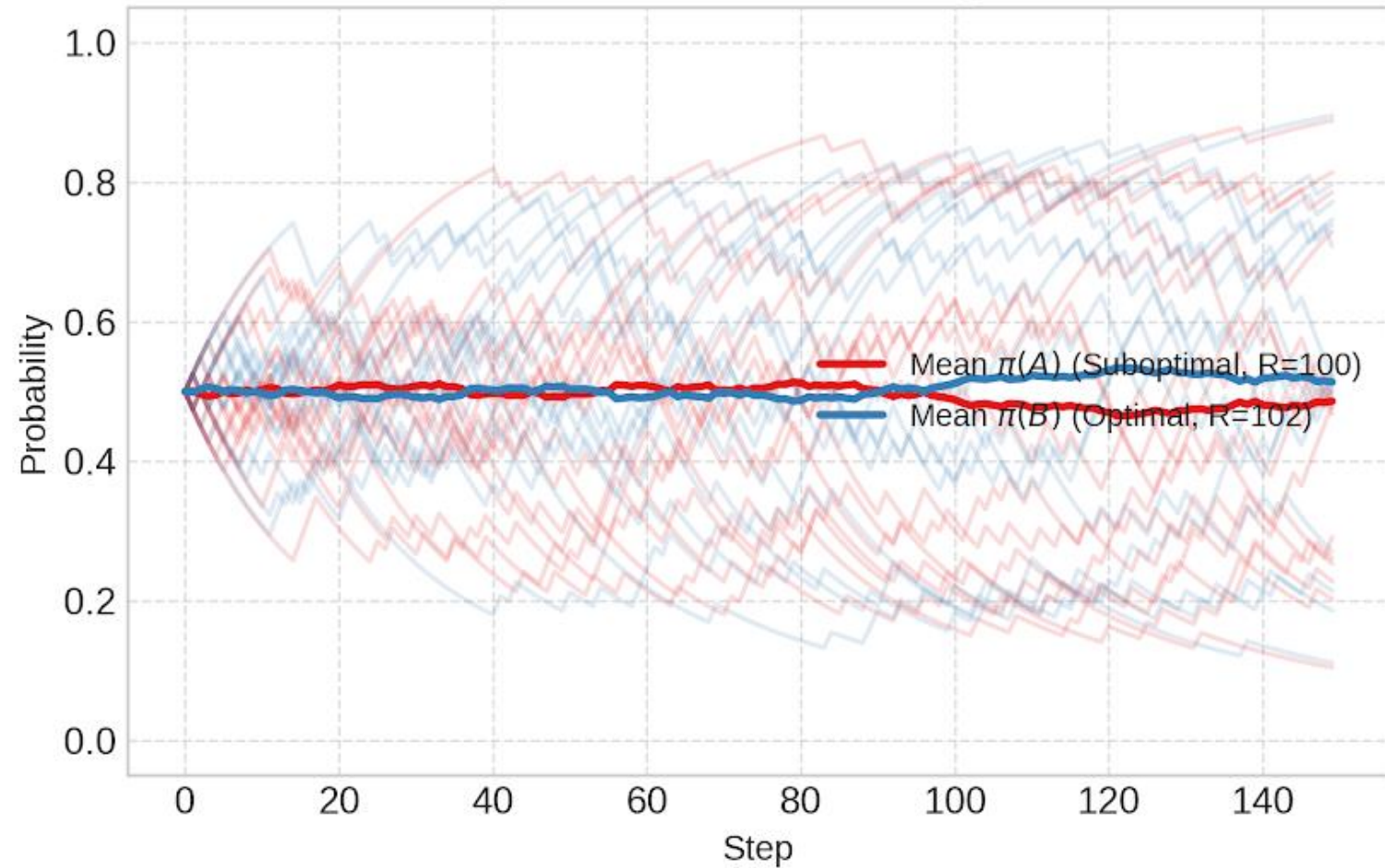
- $r(A) = 100$ $r(B) = 102$ (*unknown*)
- $\pi_\theta(A) = 0.5$
- $\pi_\theta(B) = 0.5$

- $\nabla_\theta J_A(\theta) = \nabla_\theta \log \pi(A) 100 = -\nabla_\theta \log \pi(B) 100$
- $\nabla_\theta J_B(\theta) = \nabla_\theta \log \pi(B) 102 = -\nabla_\theta \log \pi(A) 102$

Example: Bandits



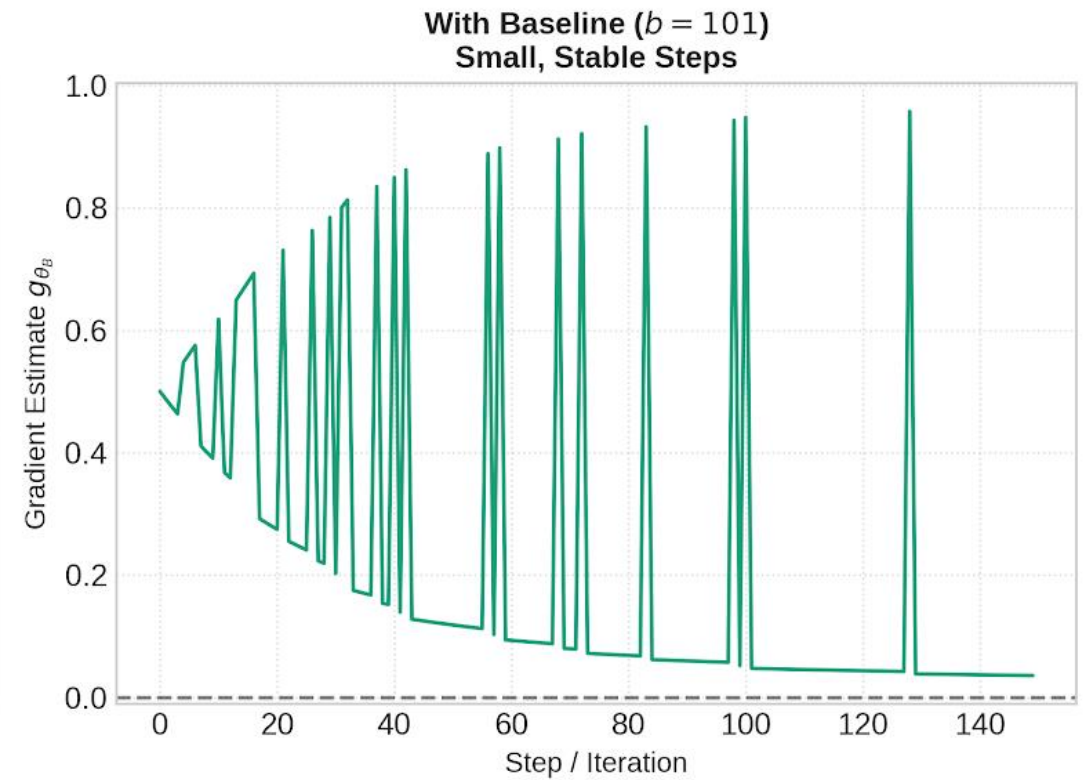
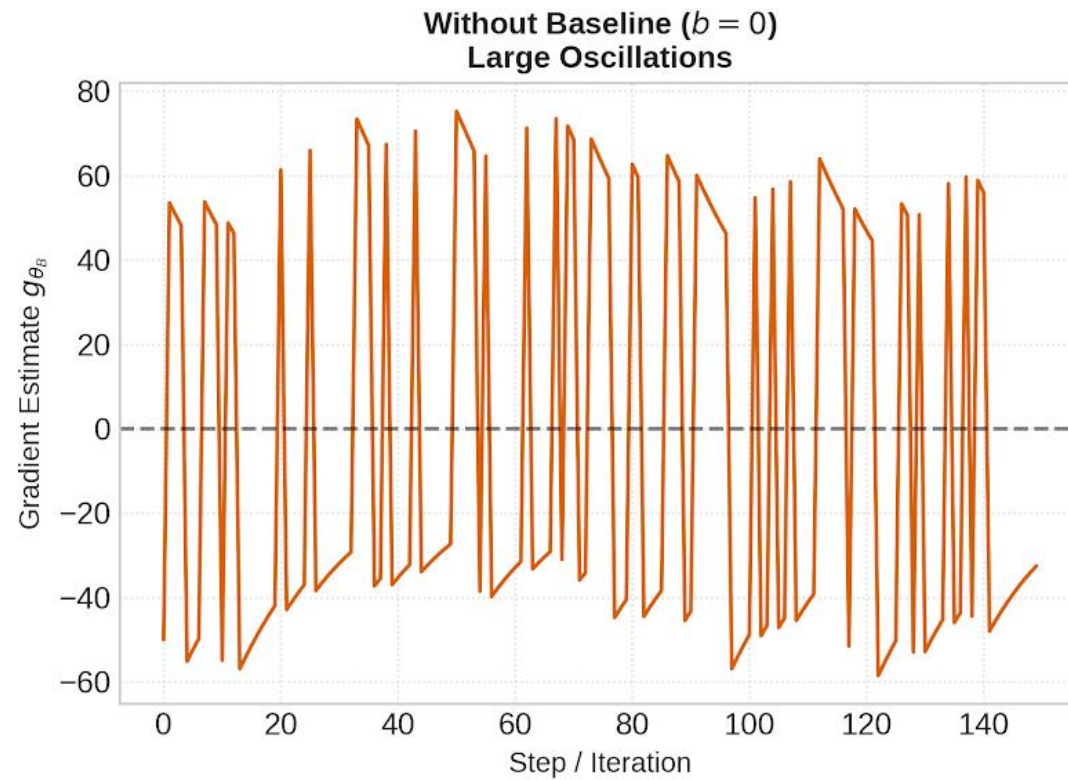
Example: Bandits



Example: Bandits

- Idea: Change the rewards.
- $r(A) = 100 - b$ $r(B) = 102 - b$
- In particular, let's use as b the average reward $b = r_{avg} = 101$
 - Since I don't know the rewards at the beginning, I can use a running estimate of the average.
- $\nabla_{\theta} J_A(\theta) = -\nabla_{\theta} \log \pi(A)$
- $\nabla_{\theta} J_B(\theta) = \nabla_{\theta} \log \pi(B)$

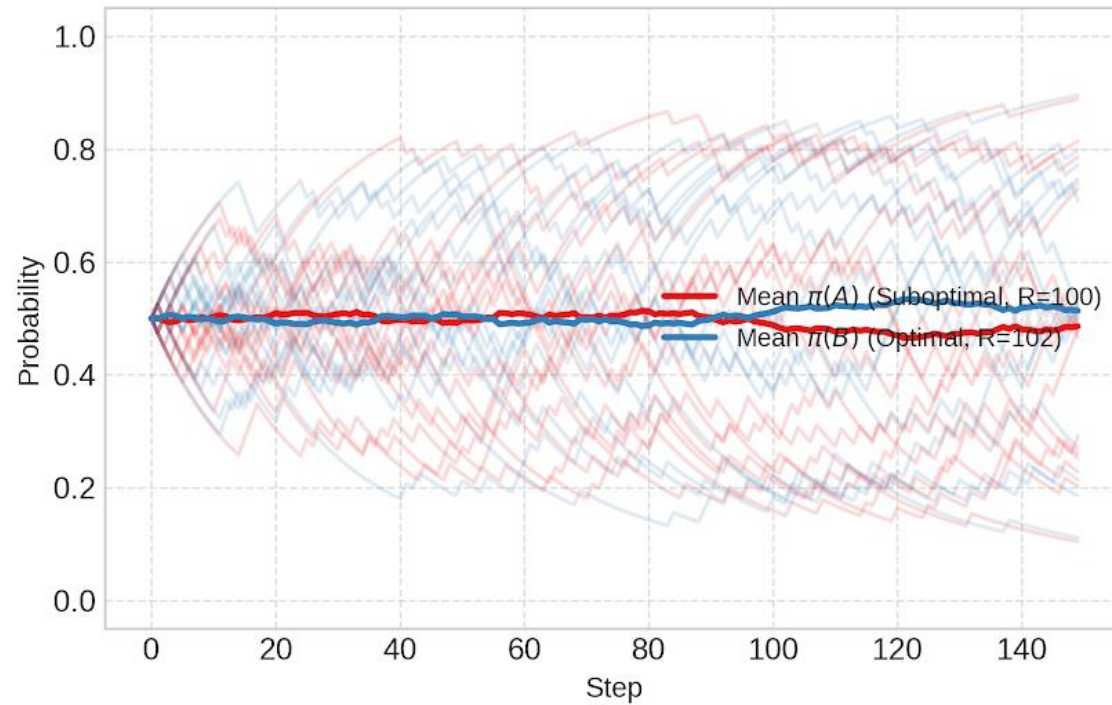
Example: Bandits



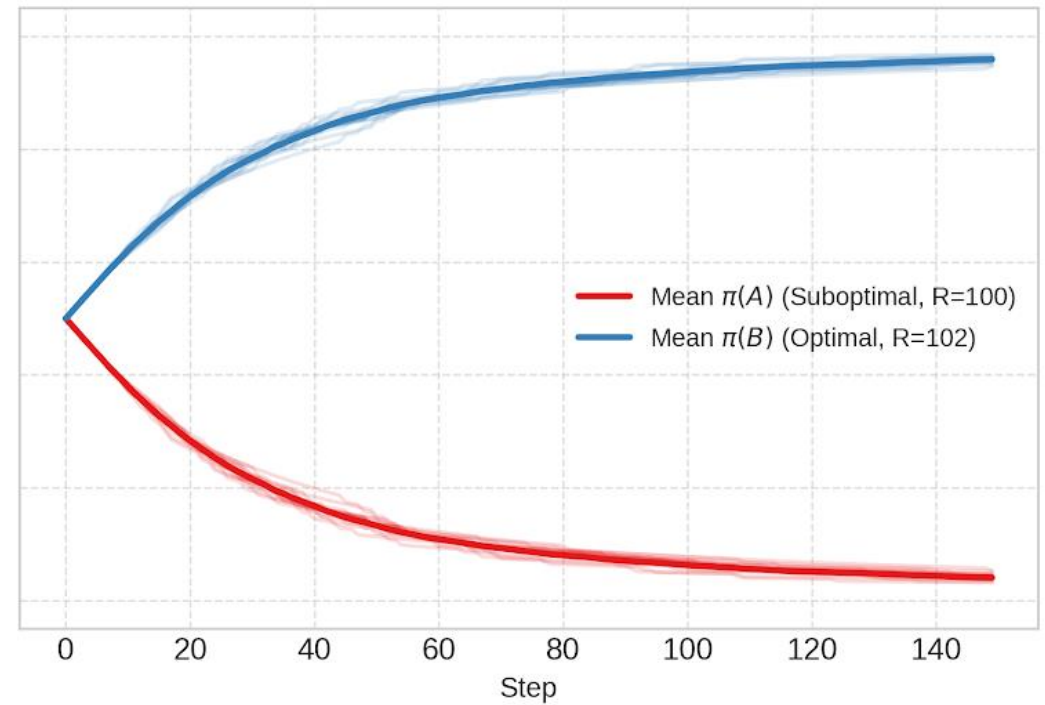
Example: Bandits



Without Baseline ($b = 0$)
High Variance / Slow Convergence



With Baseline ($b = 101$)
Low Variance / Fast Convergence



Today's Agenda

- Show that any baseline $b(s_t)$ leaves the policy gradient unchanged.
- Show that the policy value function $V^\pi(s_t)$ is (close to) the best possible baseline to decrease variance.

Policy Gradient Theorem: Baselines

- We have shown that for episodic MDPs:

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \mathbb{E}_{\tau \sim p(\tau; \theta)} [R(\tau)] = \mathbb{E}_{\tau \sim p(\tau; \theta)} \left[\sum_{t=0}^{T-1} (\nabla_{\theta} \log \pi(a_t | s_t, \theta) \textcolor{red}{G}_t) \right]$$

- Next, we will show that, $\forall b(s)$:

$$\begin{aligned} \nabla_{\theta} J(\theta) &= \mathbb{E}_{\tau \sim p(\tau; \theta)} \left[\sum_{t=0}^{T-1} (\nabla_{\theta} \log \pi(a_t | s_t, \theta) \textcolor{red}{G}_t) \right] \\ &= \mathbb{E}_{\tau \sim p(\tau; \theta)} \left[\sum_{t=0}^{T-1} \left(\nabla_{\theta} \log \pi(a_t | s_t, \theta) (\textcolor{red}{G}_t - b(s_t)) \right) \right] \end{aligned}$$

Baselines Don't Contribute to The Gradient

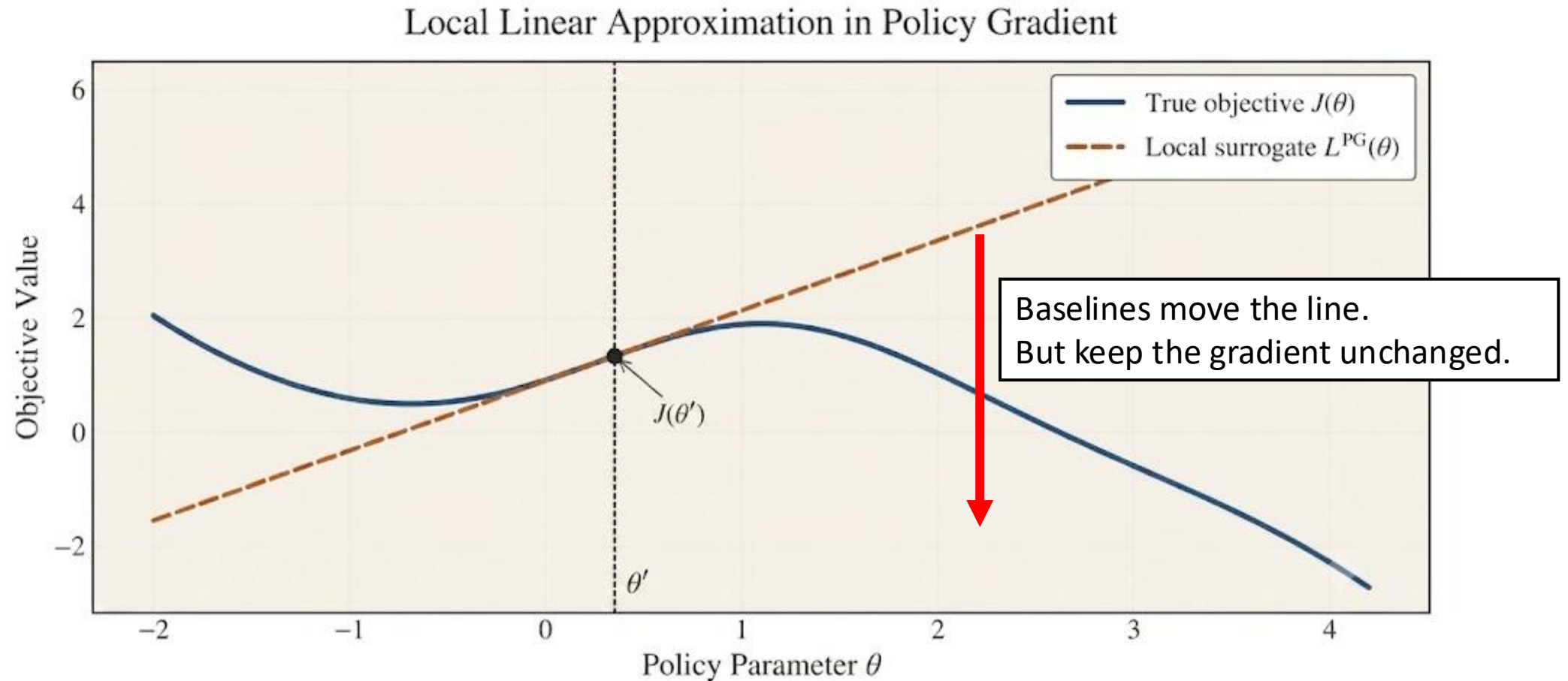
- $\mathbb{E}_{\tau \sim p(\tau; \theta)} [\sum_{t=0}^{T-1} (\nabla_{\theta} \log \pi(a_t | s_t, \theta) G_t)] - \mathbb{E}_{\tau \sim p(\tau; \theta)} [\sum_{t=0}^{T-1} (\nabla_{\theta} \log \pi(a_t | s_t, \theta) b(s_t))]$
- Next, we'll show that

$$\mathbb{E}_{\tau \sim p(\tau; \theta)} [\nabla_{\theta} \log \pi(a_t | s_t, \theta) b(s_t)] = 0$$
- $\mathbb{E}_{s_{0:t}, a_{0:t-1}} [\mathbb{E}_{s_{t+1:T-1}, a_{t:T-1}} [\nabla_{\theta} \log \pi(a_t | s_t, \theta) b(s_t)]]$ (law of total expectation)
- $\mathbb{E}_{s_{0:t}, a_{0:t-1}} [b(s_t) \mathbb{E}_{s_{t+1:T-1}, a_{t:T-1}} [\nabla_{\theta} \log \pi(a_t | s_t, \theta)]]$ ($b(s_t)$ independent of a_t)
- $\mathbb{E}_{s_{0:t}, a_{0:t-1}} [b(s_t) \mathbb{E}_{a_t} [\nabla_{\theta} \log \pi(a_t | s_t, \theta)]]$ (π independent of future)
- $\mathbb{E}_{s_{0:t}, a_{0:t-1}} \left[b(s_t) \mathbb{E}_{a_t} \left[\frac{\nabla_{\theta} \pi(a_t | s_t, \theta)}{\pi(a_t | s_t, \theta)} \right] \right]$ (expand derivative)

Baselines Don't Contribute to The Gradient

- $\mathbb{E}_{s_{0:t}, a_{0:t-1}} \left[b(s_t) \mathbb{E}_{a_t} \left[\frac{\nabla_{\theta} \pi(a_t | s_t, \theta)}{\pi(a_t | s_t, \theta)} \right] \right]$ (expand derivative)
- $\mathbb{E}_{s_{0:t}, a_{0:t-1}} \left[b(s_t) \sum_{a_t} \pi(a_t | s_t, \theta) \frac{\nabla_{\theta} \pi(a_t | s_t, \theta)}{\pi(a_t | s_t, \theta)} \right]$ (expand expectation)
- $\mathbb{E}_{s_{0:t}, a_{0:t-1}} \left[b(s_t) \nabla_{\theta} \sum_{a_t} \pi(a_t | s_t, \theta) \right]$ (take gradient out)
- $\mathbb{E}_{s_{0:t}, a_{0:t-1}} [b(s_t) \nabla_{\theta} 1] = 0$
- This is sufficient to show that $\mathbb{E}_{\tau \sim p(\tau; \theta)} \left[\sum_{t=0}^{T-1} \left(\nabla_{\theta} \log \pi(a_t | s_t, \theta) b(s_t) \right) \right] = 0$
 - Baselines do not contribute to the gradient!

Policy Gradients with Baselines



Baselines

- If they don't change the gradient, why should I use them?
- As we will show next, baselines are helpful for **reducing the policy gradient's variance**.
- Next, we will find the baseline $b(st)$ that *minimizes* variance.

Baselines for Variance Reduction

- $\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim p(\tau; \theta)} \left[\sum_{t=0}^{T-1} \left(\nabla_{\theta} \log \pi(a_t | s_t, \theta) (G_t - b(s_t)) \right) \right]$
- $\mathbb{E}_{\tau_t = (s_t, a_t, \dots, s_{T-1}, a_{T-1}) \sim \pi} \left[\nabla_{\theta} \log \pi(a_t | s_t, \theta) (G_t - b(s_t)) \right]$
- $l_t = \nabla_{\theta} \log \pi(a_t | s_t, \theta) \quad b_t = b(s_t)$
- $\nabla_{\theta} J^t(\theta) = \mathbb{E}_{\tau_t \sim \pi} [l_t (G_t - b_t)]$
- Let's focus on the variance of the random variable $l_t (G_t - b_t)$

Baselines for Variance Reduction

- $\sigma = \text{Var}_{\tau_t \sim \pi}[l_t(G_t - b_t)]$
- What b_t minimizes σ ?
- $\mathbb{E}_{\tau_t \sim \pi}[l_t^2(G_t - b_t)^2] - (\mathbb{E}_{\tau_t \sim \pi}[l_t(G_t - b_t)])^2$ (variance def.)
- $\mathbb{E}_{\tau_t \sim \pi}[l_t^2(G_t - b_t)^2] - (\mathbb{E}_{\tau_t \sim \pi}[l_t G_t] + \mathbb{E}_{\tau_t \sim \pi}[l_t b_t])^2$
- $\mathbb{E}_{\tau_t \sim \pi}[l_t^2(G_t - b_t)^2] - (\mathbb{E}_{\tau_t \sim \pi}[l_t G_t])^2$ (as proved earlier)
- $\mathbb{E}_{\tau_t \sim \pi}[l_t^2 G_t^2] + \mathbb{E}_{\tau_t \sim \pi}[l_t^2 \textcolor{red}{b}_t^2] - 2\mathbb{E}_{\tau_t \sim \pi}[l_t^2 G_t \textcolor{red}{b}_t] - (\mathbb{E}_{\tau_t \sim \pi}[l_t G_t])^2$

This is a quadratic form (parabola) with respect to b_t !

Baselines for Variance Reduction

- $\sigma = \mathbb{E}_{\tau_t \sim \pi}[l_t^2 G_t^2] + \mathbb{E}_{\tau_t \sim \pi}[l_t^2 \textcolor{red}{b}_t^2] - 2\mathbb{E}_{\tau_t \sim \pi}[l_t^2 G_t \textcolor{red}{b}_t] - (\mathbb{E}_{\tau_t \sim \pi}[l_t G_t])^2$
- Let's compute the functional derivative $\frac{\delta \sigma}{\delta b_t}$
- At any state s_t :
- $\frac{\delta \sigma}{\delta b_t} = 2\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi}[l_t^2 b_t] - 2\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi}[l_t^2 G_t]$
- Since it is a quadratic with a positive coefficient, b_t^* minimizing σ has
- $\frac{d\sigma}{db_t} \big|_{b_t^*} = 0$

Baselines for Variance Reduction

- $\frac{d\sigma}{db_t} \big|_{b_t^*} = 0$
- $2\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi} [l_t^2 b_t^*] - 2\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi} [l_t^2 G_t] = 0$
- $b_t^* = \frac{\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi} [l_t^2 G_t]}{\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi} [l_t^2]}$
- Expanding the terms:
- $b^*(s_t) = \frac{\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi} [\|\nabla_{\theta} \log \pi(a_t | s_t)\|^2 G_t]}{\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi} [\|\nabla_{\theta} \log \pi(a_t | s_t)\|^2]}$
- $b^*(s_t) = \frac{\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi} [\|\nabla_{\theta} \log \pi(a_t | s_t)\|^2 G_t]}{\mathbb{E}_{a_t \sim \pi} [\|\nabla_{\theta} \log \pi(a_t | s_t)\|^2]}$ (a_t independent to future)

The optimal baseline

- $b^*(s_t) = \frac{\mathbb{E}_{a_t, s_{t+1}, a_{t+1}, \dots \sim \pi} [\|\nabla_{\theta} \log \pi(a_t | s_t)\|^2 G_t]}{\mathbb{E}_{a_t \sim \pi} [\|\nabla_{\theta} \log \pi(a_t | s_t)\|^2]}$
- $b^*(s_t) = \frac{\mathbb{E}_{a_t \sim \pi} [\|\nabla_{\theta} \log \pi(a_t | s_t)\|^2 \mathbb{E}_{s_{t+1}, a_{t+1}, \dots | a_t \sim \pi} [G_t]]}{\mathbb{E}_{a_t \sim \pi} [\|\nabla_{\theta} \log \pi(a_t | s_t)\|^2]}$ (law of total expect.)
- What is $\mathbb{E}_{s_{t+1}, a_{t+1}, \dots | a_t \sim \pi} [G_t] = \mathbb{E}_{s_{t+1}, a_{t+1}, \dots | a_t \sim \pi} [\sum_{t'=t}^{T-1} \gamma^{t-t'} r(s_{t'}, a_{t'})]$?
- In words: Expected return if taking action a_t and then following π .

The optimal baseline

- $\mathbb{E}_{s_{t+1}, a_{t+1}, \dots | a_t \sim \pi} [G_t] = Q^\pi(s_t, a_t)$
- $$b^*(s_t) = \frac{\mathbb{E}_{a_t \sim \pi} [\|\nabla_\theta \log \pi(a_t | s_t)\|^2 Q^\pi(s_t, a_t)]}{\mathbb{E}_{a_t \sim \pi} [\|\nabla_\theta \log \pi(a_t | s_t)\|^2]}$$
- Assume discrete actions:
- $$b^*(s_t) = \frac{\sum_{a_t \in A} \pi(a_t | s_t) \|\nabla_\theta \log \pi(a_t | s_t)\|^2 Q^\pi(s_t, a_t)}{\sum_{a_t \in A} \pi(a_t | s_t) \|\nabla_\theta \log \pi(a_t | s_t)\|^2}$$

The optimal baseline

- Assume discrete actions:
- $$b^*(s_t) = \frac{\sum_{a_t \in A} \pi(a_t|s_t) \|\nabla_{\theta} \log \pi(a_t|s_t)\|^2 Q^{\pi}(s_t, a_t)}{\sum_{a_t \in A} \pi(a_t|s_t) \|\nabla_{\theta} \log \pi(a_t|s_t)\|^2}$$
- The optimal baseline at s_t is the average of the Q values at s_t
 - Weighted by the product of the probability of actions.
 - And the squared norm of the log-probability of the action.
- Expensive to compute: need the gradient at every possible action.

A close-to-optimal baseline

- Assume that the gradient is similar for all actions :

- $$b^*(s_t) = \frac{\sum_{a_t \in A} \pi(a_t|s_t) \cancel{\|\nabla_{\theta} \log \pi(a_t|s_t)\|^2} Q^{\pi}(s_t, a_t)}{\sum_{a_t \in A} \pi(a_t|s_t) \cancel{\|\nabla_{\theta} \log \pi(a_t|s_t)\|^2}}$$

- $$b^*(s_t) = \sum_{a_t \in A} \pi(a_t|s_t) Q^{\pi}(s_t, a_t)$$

- What is this $b^*(s_t)$ equal to?

A close-to-optimal baseline

- Assume that the gradient is similar for all actions :

- $$b^*(s_t) = \frac{\sum_{a_t \in A} \pi(a_t|s_t) \cancel{\|\nabla_{\theta} \log \pi(a_t|s_t)\|^2} Q^{\pi}(s_t, a_t)}{\sum_{a_t \in A} \pi(a_t|s_t) \cancel{\|\nabla_{\theta} \log \pi(a_t|s_t)\|^2}}$$

- $$b^*(s_t) = \sum_{a_t \in A} \pi(a_t|s_t) Q^{\pi}(s_t, a_t)$$

- $b^*(s_t) = V^{\pi}(s_t)$. The value function is a close to optimal baseline!

The Advantage Function

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim p(\tau; \theta)} \left[\sum_{t=0}^{T-1} \left(\nabla_{\theta} \log \pi(a_t | s_t, \theta) \left(G_t - V^{\pi}(s_t) \right) \right) \right]$$

- The term in red is a sample of the *advantage function*:

$$A^{\pi}(s_t, a_t) = Q^{\pi}(s_t, a_t) - V^{\pi}(s_t)$$

- This function quantifies the “advantage” of action a_t with respect to the policy’s actions, on average.

Example: Bandits

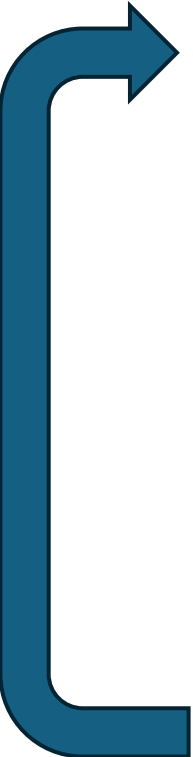
Choose

A

B

- $r(A) = 100$ $r(B) = 102$ (*unknown*)
- $\pi_\theta(A) = 0.5$
- $\pi_\theta(B) = 0.5$
- $V^\pi = 101$
- $A^\pi(A) = 100 - 101 = -1$
- $A^\pi(B) = 102 - 101 = 1$

Advantage Actor-Critic (A2C)

- 
- Init θ at random
 - Run policy $\pi_{\theta}(a_t|s_t)$ and collect an episode in the environment.
 - For each transition $(s_t, a_t, r_t, s_{t'})$
 - Compute total return $G_t = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$
 - Compute the sample advantage $A_t = G_t - V^{\pi}(s_t)$
 - Compute the policy gradient estimate:
 - $\hat{g} = \frac{1}{T-1} \sum_t \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) A_t$
 - Update policy parameters
 - $\theta \leftarrow \theta + \alpha \hat{g}$

• A2C is the foundational policy-gradient algorithm

Value Function Approximation (Refresher)

- The baseline requires the true value function $V^\pi(s)$.
- But, in practice, we learn an approximation of $\hat{V}(s) \approx V^\pi(s)$
- Using tabular methods if the state space is discrete and “small”
- Or parametric approximations if the state space is continuous or very large

$$\hat{V}(s, \mathbf{w}) \approx V^\pi(s)$$

Tabular Value Function Approximation

- Initialize a table $V(s)$ with $|S|$ entries.
- Update each entry in the table according to the observed returns.
- MC: the target is the sample return G_t

$$V(s_t) \leftarrow V(s_t) + \alpha(G_t - V(s_t))$$

- $TD(0)$: the target is the TD target $r_{t+1} + \gamma V(s_{t+1})$
$$V(s_t) \leftarrow V(s_t) + \alpha(r_{t+1} + \gamma V(s_{t+1}) - V(s_t))$$

- $TD(\lambda)$: the target is the λ -return G_t^λ
$$V(s_t) \leftarrow V(s_t) + \alpha(G_t^\lambda - V(s_t))$$

Parametric Value Function Approximation

- We learn parameters $\mathbf{w}$ such that $\hat{V}(s, \mathbf{w}) \approx V^\pi(s)$

- For MC, the target is the sample return G_t

$$\Delta \mathbf{w} = \alpha \left(G_t - \hat{V}(s, \mathbf{w}) \right) \nabla_{\mathbf{w}} \hat{V}(s, \mathbf{w})$$

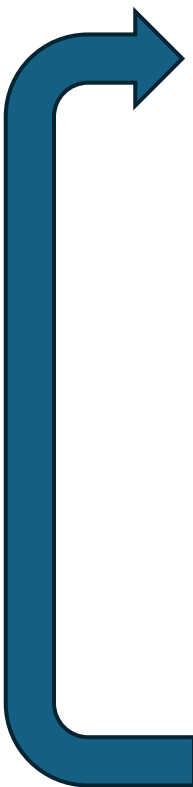
- For $TD(0)$, the target is the TD target $r_{t+1} + \gamma \hat{V}(s_{t+1}, \mathbf{w})$

$$\Delta \mathbf{w} = \alpha \left(r_{t+1} + \gamma \hat{V}(s_{t+1}, \mathbf{w}) - \hat{V}(s, \mathbf{w}) \right) \nabla_{\mathbf{w}} \hat{V}(s, \mathbf{w})$$

- For $TD(\lambda)$, the target is the λ -return G_t^λ

$$\Delta \mathbf{w} = \alpha \left(G_t^\lambda - \hat{V}(s, \mathbf{w}) \right) \nabla_{\mathbf{w}} \hat{V}(s, \mathbf{w})$$

Advantage Actor-Critic (Revised)

- 
- Init θ at random
 - Run policy $\pi_{\theta}(a_t|s_t)$ and collect an episode in the environment.
 - For each transition $(s_t, a_t, r_t, s_{t'})$
 - Compute total return $G_t = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$
 - Compute the advantage $A_t = G_t - \hat{V}(s_t)$
 - Compute the policy gradient estimate:
 - $\hat{g} = \frac{1}{T-1} \sum_t \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) A_t$
 - Update policy parameters
 - $\theta \leftarrow \theta + \alpha \hat{g}$
 - Update the approximation of the value function

Policy Gradient Theorem

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim p(\tau; \theta)} \left[\sum_{t=0}^{T-1} \left(\nabla_{\theta} \log \pi(a_t | s_t, \theta) \left(G_t - \hat{V}(s_t) \right) \right) \right]$$

- Instead of using the full return G_t in the gradient
- We could use a bootstrapped estimate:
 - $TD(0)$: $G_t \sim r_t + \gamma V(s_{t+1})$
 - $TD(n)$: $G_t \sim r_t + \gamma r_{t+1} + \dots + \gamma^{n-1} V(s_{t+n})$
 - $TD(\lambda)$: $G_t \sim G_t^{\lambda}$
- The bootstrapped estimates can help to further decrease variance.
 - At the cost of some bias.
- This is distinct from the bootstrapped estimates for $\hat{V}$
 - But almost always the same technique is used.

An interesting Special Case: $TD(\lambda)$ in PG

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim p(\tau; \theta)} \left[\sum_{t=0}^{T-1} \left(\nabla_{\theta} \log \pi(a_t | s_t, \theta) \left(G_t^{\lambda} - \hat{V}(s_t) \right) \right) \right]$$

- Using telescoping, we can write

$$G_t^{\lambda} - V^{\pi}(s_t) = \sum_{t'=t}^{T-1} (\gamma \lambda)^{(t'-t)} \delta_{t'}$$

- Where $\delta_{t'} = r_{t'} + \gamma \hat{V}(s_{t'+1}) - \hat{V}(s_{t'})$
 - The TD error at time t'
- This is called “Generalized Advantage Estimation”
 - Can be implemented very efficiently.
 - And reused when updating the value function approximation

Advantage Actor-Critic (Revised)

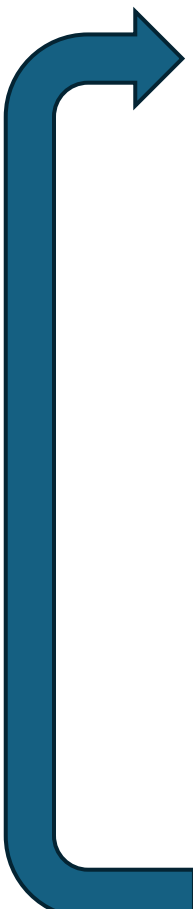
- Init θ at random
- Run policy $\pi_{\theta}(a_t|s_t)$ and collect an episode in the environment.
- For each transition $(s_t, a_t, r_t, s_{t'})$
 - Compute the (bootstrapped) return G_t^{λ}
 - Compute the advantage $A_t = G_t^{\lambda} - \hat{V}(s_t)$

} Can do these two simultaneously with Generalized Advantage Estimation
- Compute the policy gradient estimate:
 - $\hat{g} = \frac{1}{T-1} \sum_t \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) A_t$
- Update policy parameters
 - $\theta \leftarrow \theta + \alpha \hat{g}$
- Update the approximation of the value function

Going Off-Policy

- A2C is a *on-policy* algorithm
 - Only the data generated by the current policy is used in the updates
- Can we compute the policy gradient using trajectories of different policies?
- Yes! This will make A2C more sample-efficient.
- Many variants are possible (A3C, TRPO, PPO, etc.).

Practical Implementation

- 
- Init θ at random
 - Run policy $\pi_{\theta}(a_t|s_t)$ and collect an episode in the environment.
 - For each transition $(s_t, a_t, r_t, s_{t'})$
 - Compute the (bootstrapped) return G_t^{λ}
 - Compute the advantage $A_t = G_t^{\lambda} - \hat{V}(s_t)$
 - Compute the policy gradient loss:
 - $L_{pg} = \frac{1}{T-1} \sum_t \log \pi_{\theta}(a_t|s_t) A_t$
 - Use auto-differentiation libraries to compute the policy gradient
 - $\hat{g} = \text{AutoDiff}(L_{pg}; \theta)$
 - Update policy parameters
 - $\theta \leftarrow \theta + \alpha \hat{g}$
 - Update the approximation of the value function